

Task 1: What the Data Looks Like and the Gaps I Found

1. What the Data Looks Like

- **Data Scale & Quality:** Large, clean sEMG time-series dataset. 18,011,280 raw samples across 10 subjects (12 channels, 2000 Hz), with 0 missing values and 0 duplicate frames.
- **Signal Volatility:** Microvolt-level signal with high kurtosis (up to ~711 on Channel 3) and frequent high-amplitude contraction spikes (9%–27% IQR outliers per channel).
- **Feature Relationships:** Strong correlation ($r > 0.85$) between MAV and RMS on the same channel, 31 redundant feature pairs detected while Zero Crossing carries more independent frequency information.
- **Manifold Complexity:** Linear PCA leaves gesture classes heavily overlapping; the non-linear UMAP projection shows clearer clustering, isolating Rest from active gesture states.

2. Identified Gaps & Modeling Challenges

- **Class Distribution Gap:** Rest (Class 0) dominates 50%–59% of each subject's timeline, creating an inflated ~55% baseline-accuracy trap for naive classifiers.
- **Near-Constant Channel Gap:** 4 of 12 raw channels (Ch2, Ch5, Ch6, Ch11) show near-zero variance, hinting at limited discriminative signal from those electrode sites.
- **Feature Redundancy Gap:** 31 feature pairs exceed $r > 0.85$ (mostly MAV vs. RMS per channel), risking multicollinearity for linear baseline models.
- **Pipeline Mitigation Gap:** Needs robust scaling (to preserve genuine contraction spikes rather than strip them as noise), redundancy-aware feature handling or spatial-graph modeling (GNN), and imbalance-aware validation (Macro-F1) given the severe Rest bias.